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Roll Number: S24CSEU1460

Batch: 32

Laboratory session date:25/09/26

2) https://github.com/Mayankithub/ML_OPS_L1

3)config.yaml

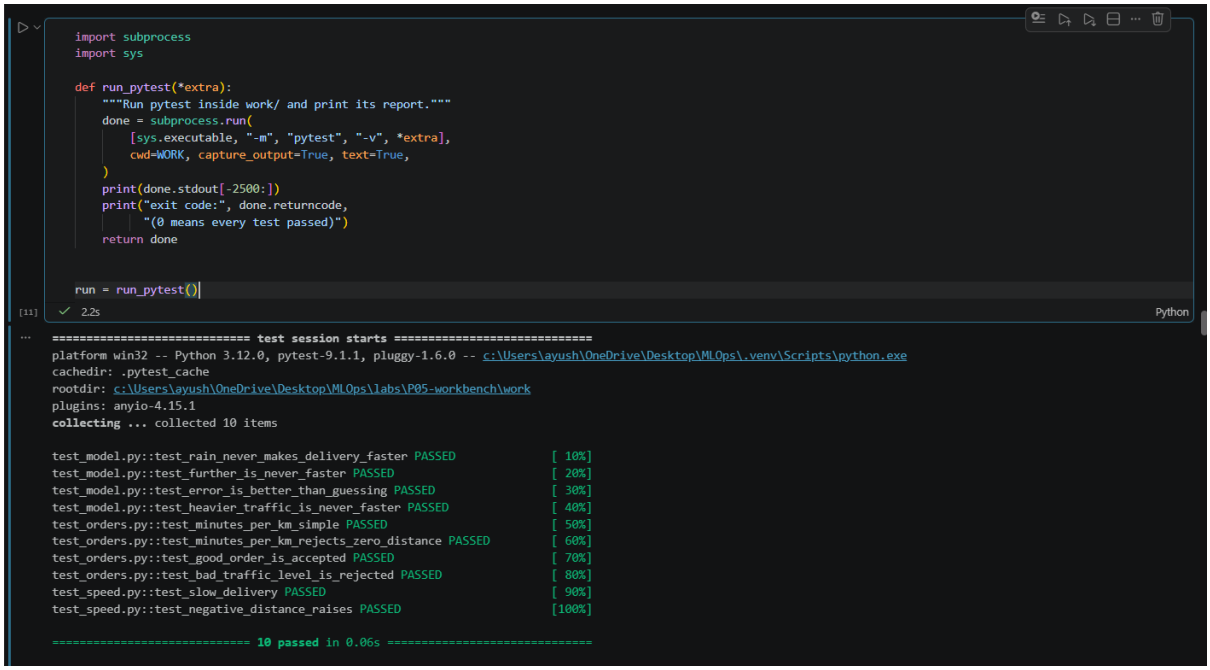
```
# Settings for the delivery-time model.
# Change values here, never in the code.

data:
  path: ../data/delivery_times.csv
  features:
    - distance_km
    - prep_time_min
    - traffic_level
    - rain
  target: delivery_min

split:
  test_size: 0.2
  seed: 42

model:
  name: linear_regression
```

4)



```python
import subprocess
import sys

def run_pytest(*extra):
    """Run pytest inside work/ and print its report."""
    done = subprocess.run(
        [sys.executable, "-m", "pytest", "-v", *extra],
        cwd=WORK, capture_output=True, text=True,
    )
    print(done.stdout[-2500:])
    print("exit code:", done.returncode,
          "(0 means every test passed)")
    return done


run = run_pytest()
```

```
[11] ✓ 2.2s                                                    Python

============================= test session starts =============================
platform win32 -- Python 3.12.0, pytest-9.1.1, pluggy-1.6.0 -- c:\Users\ayush\OneDrive\Desktop\MLOps\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: c:\Users\ayush\OneDrive\Desktop\MLOps\labs\P05-workbench\work
plugins: anyio-4.15.1
collecting ... collected 10 items

test_model.py::test_rain_never_makes_delivery_faster PASSED              [ 10%]
test_model.py::test_further_is_never_faster PASSED                       [ 20%]
test_model.py::test_error_is_better_than_guessing PASSED                 [ 30%]
test_model.py::test_heavier_traffic_is_never_faster PASSED               [ 40%]
test_orders.py::test_minutes_per_km_simple PASSED                        [ 50%]
test_orders.py::test_minutes_per_km_rejects_zero_distance PASSED         [ 60%]
test_orders.py::test_good_order_is_accepted PASSED                       [ 70%]
test_orders.py::test_bad_traffic_level_is_rejected PASSED                [ 80%]
test_speed.py::test_slow_delivery PASSED                                 [ 90%]
test_speed.py::test_negative_distance_raises PASSED                      [100%]

============================== 10 passed in 0.06s ==============================
```

5) Before refactor: MAE = 1.92 minutes; after refactor: MAE = 1.92 minutes.

6) I used ChatGPT as a learning and debugging aid to understand the practical instructions, clarify Python/pytest concepts, troubleshoot errors, and review my implementation. I wrote, ran, tested, and verified the submitted work myself and can explain and modify the submitted code.